

CareerBridge

Student Internship & Job Application Management System

Full System Overview — All 8 Modules

Team CHAMPION

SECJ3483 Web Technology — Group Project

What We'll Cover

1. System Architecture & Tech Stack
2.  M1 Authentication & User Management
3.  M2 Job Management — Full CRUD
4.  M3 Application Tracking
5.  M4 Company Management
6.  M5 Reporting & Analytics
7.  M6 Student Profile
8.  M7 Forum / Community
9.  M8 Mock Interviews
10. Database Design & Security

Tech Stack

Frontend

- **Vue 3** — Composition API with <script setup>
- **Vite** — dev server + build tool
- **Vue Router** — SPA navigation with role guards
- **Axios** — HTTP client with JWT interceptor
- **Tailwind CSS** — utility-first styling
- **Chart.js** — analytics dashboard charts

Backend

- **PHP 8.5** — server-side language
- **Slim 4** — micro-framework for REST API
- **PDO** — prepared statements (SQL injection safe)
- **JWT** — Firebase JWT library for auth tokens
- **MySQL** — relational database (InnoDB)
- **Composer** — PHP dependency manager

System Architecture



Request flow: User clicks → Vue sends HTTP request → Vite proxy forwards to PHP → Slim routes to Controller → PDO queries MySQL → JSON response → Vue updates UI reactively.

8 Modules — Who Built What

M1 Auth & Users

JWT login, register, role management, user directory. Shared infrastructure: Database, Json, Axios, JWT middleware.

M2 Job Management

Full CRUD for job listings. Admins create/edit/deactivate. Students browse, search, filter. Soft delete.

M3 Applications

Students apply, track status, withdraw. Admins review and change status. Duplicate prevention.

M4 Companies

Admin CRUD for company profiles. Linked to jobs via foreign key. Referential integrity checks.

M5 Analytics

Admin dashboard with stat cards and Chart.js bar chart. Aggregates data from jobs + applications.

M6 Student Profile

Matric number, programme, CGPA, skills, resume. Upsert pattern (insert or update).

M7 Forum / Community

Posts with tags, comments, like toggle. Search and filter by tag.

M8 Mock Interviews

Admins create interview slots. Students book them. Admins evaluate with scores and feedback.

Database — 10 Tables

Core Tables

- **users** — id, name, email, password_hash, role
- **companies** — id, name, industry, location, description
- **jobs** — id, company_id (FK), posted_by (FK), title, type, deadline, is_active
- **applications** — id, job_id (FK), user_id (FK), cover_letter, status, UNIQUE(job_id, user_id)
- **student_profiles** — user_id (FK), matric_no, programme, cgpa, skills

Community & Interviews

- **posts** — id, user_id (FK), title, content, tag
- **comments** — id, post_id (FK), user_id (FK), content
- **post_likes** — post_id + user_id (composite PK)
- **interview_slots** — id, interviewer_id (FK), scheduled_at, is_booked
- **mock_interviews** — slot_id (FK), student_id (FK), status, feedback, score

All tables use InnoDB with foreign key constraints for referential integrity. Passwords stored as bcrypt hashes (cost 12).

Demo Credentials

Password for all accounts: Password123!

Role	Email	What to Show
Student	samin@student.utm.my	Browse jobs, apply, my applications, profile, forum, book interview
Admin	farizal@careerbridge.my	Dashboard, manage jobs, manage applications, companies, interviews
Super Admin	superadmin@careerbridge.my	Everything admin + users directory

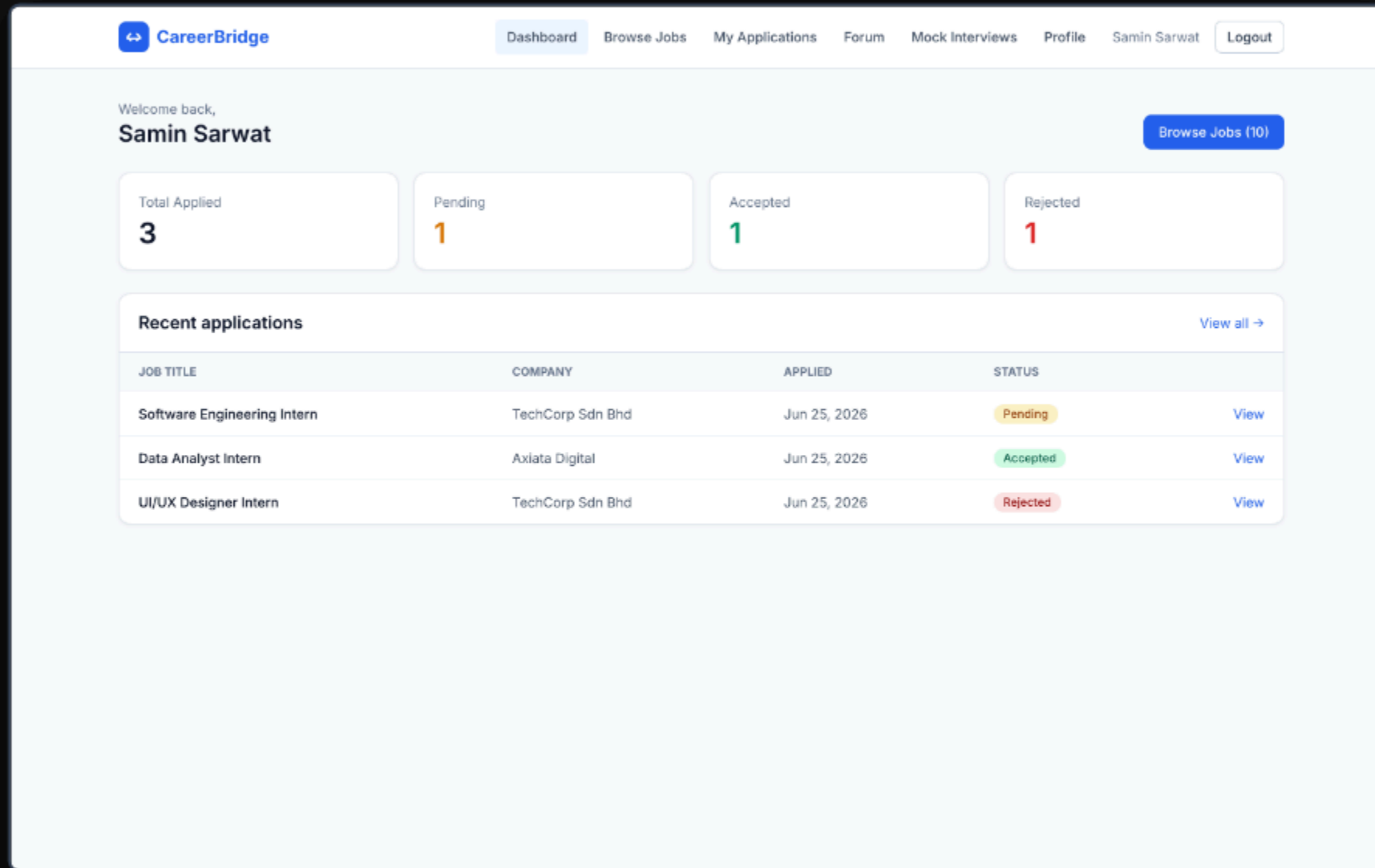
M1 Login — Entry Point



The image shows a login form for CareerBridge. It features a blue logo at the top, followed by the text "Welcome back" and "Sign in to your CareerBridge account". Below this are two input fields: "Email" with the value "you@student.utm.my" and "Password" with masked characters. A blue "Sign in" button is positioned below the password field. At the bottom, there is a link that says "New to CareerBridge? Create an account". The form is centered on a light blue background with a subtle geometric pattern.

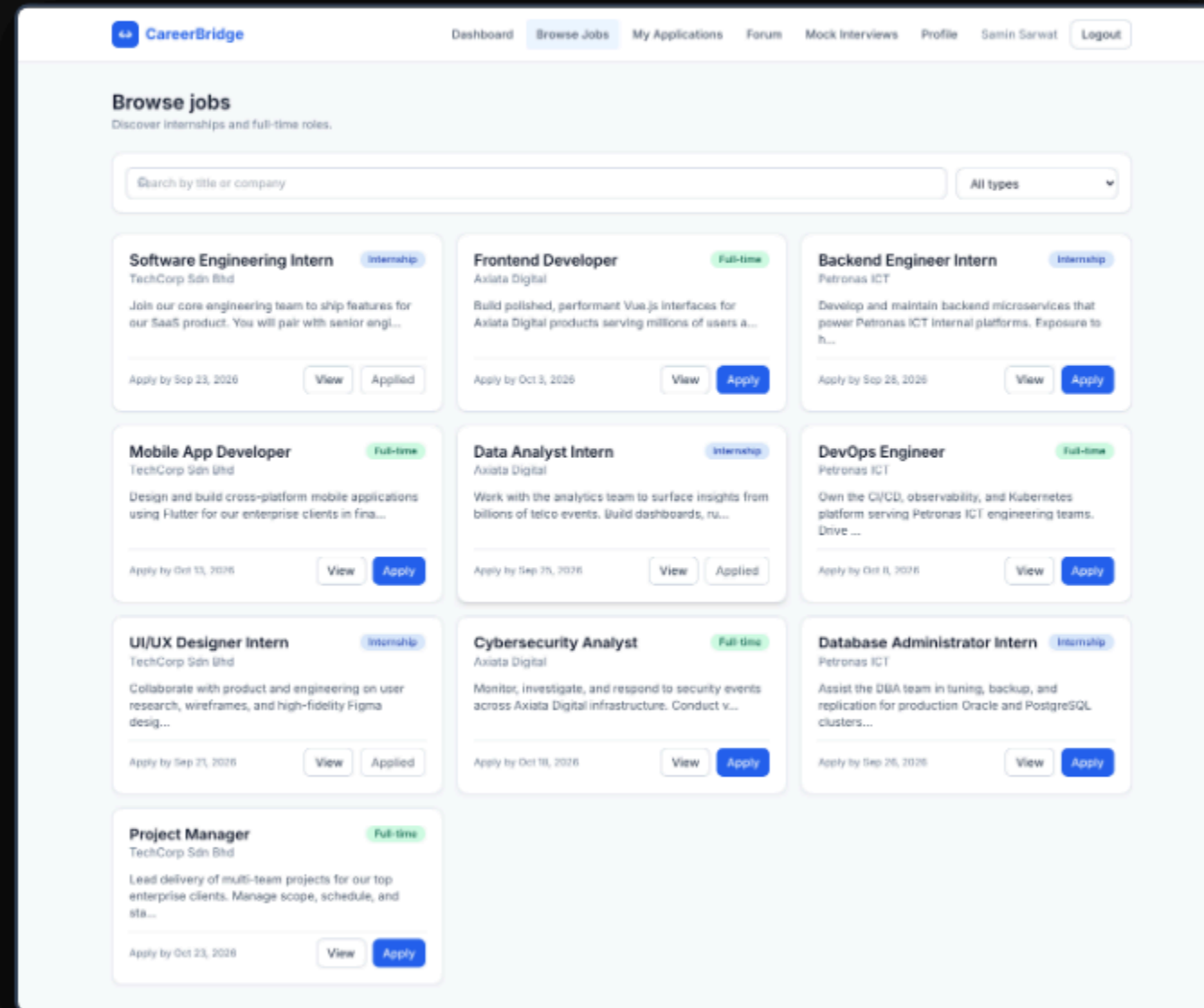
Frontend validates email format + password length. Backend verifies bcrypt hash and returns JWT token stored in localStorage.

M6 Student Dashboard



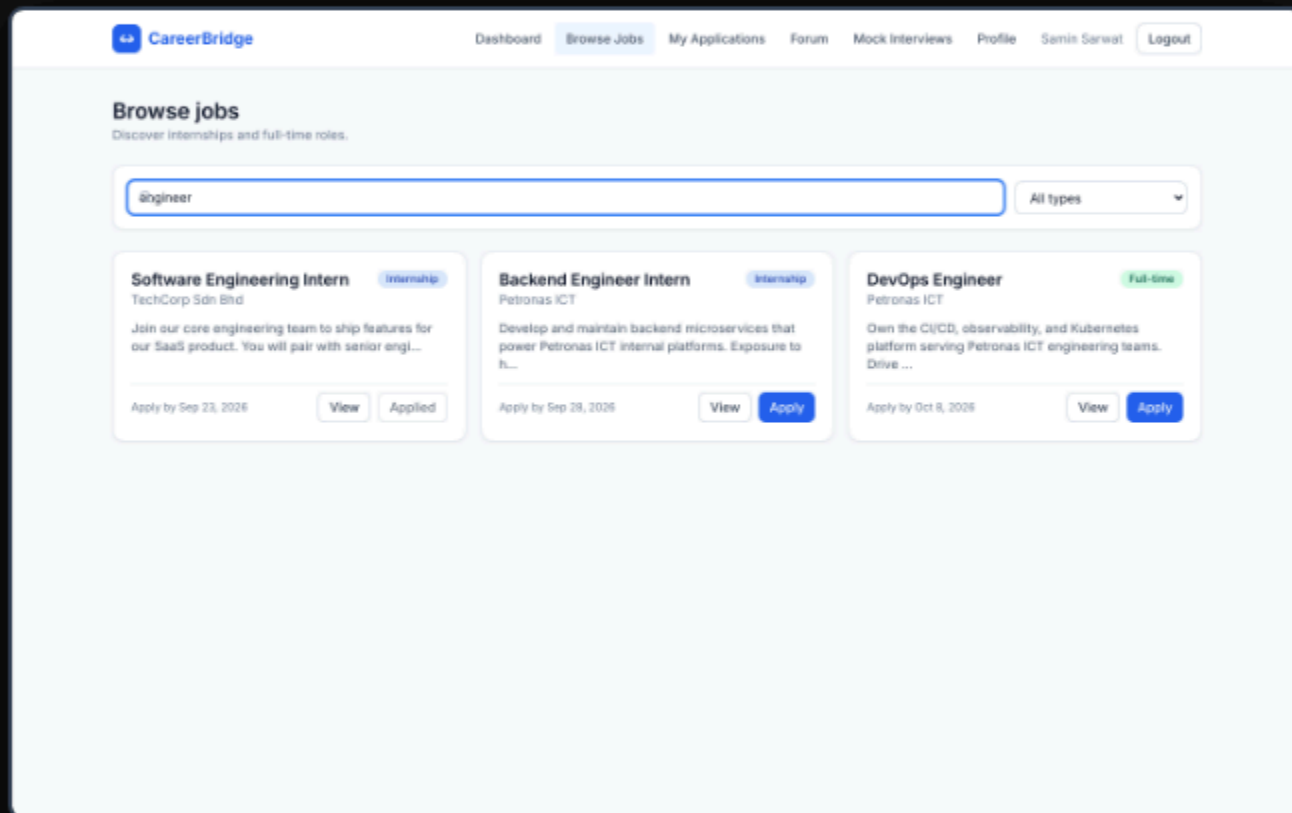
Shows application stats (total, pending, accepted, rejected) and 5 most recent applications. Data from GET /api/applications (student-scoped).

M2 Browse Jobs (Student)

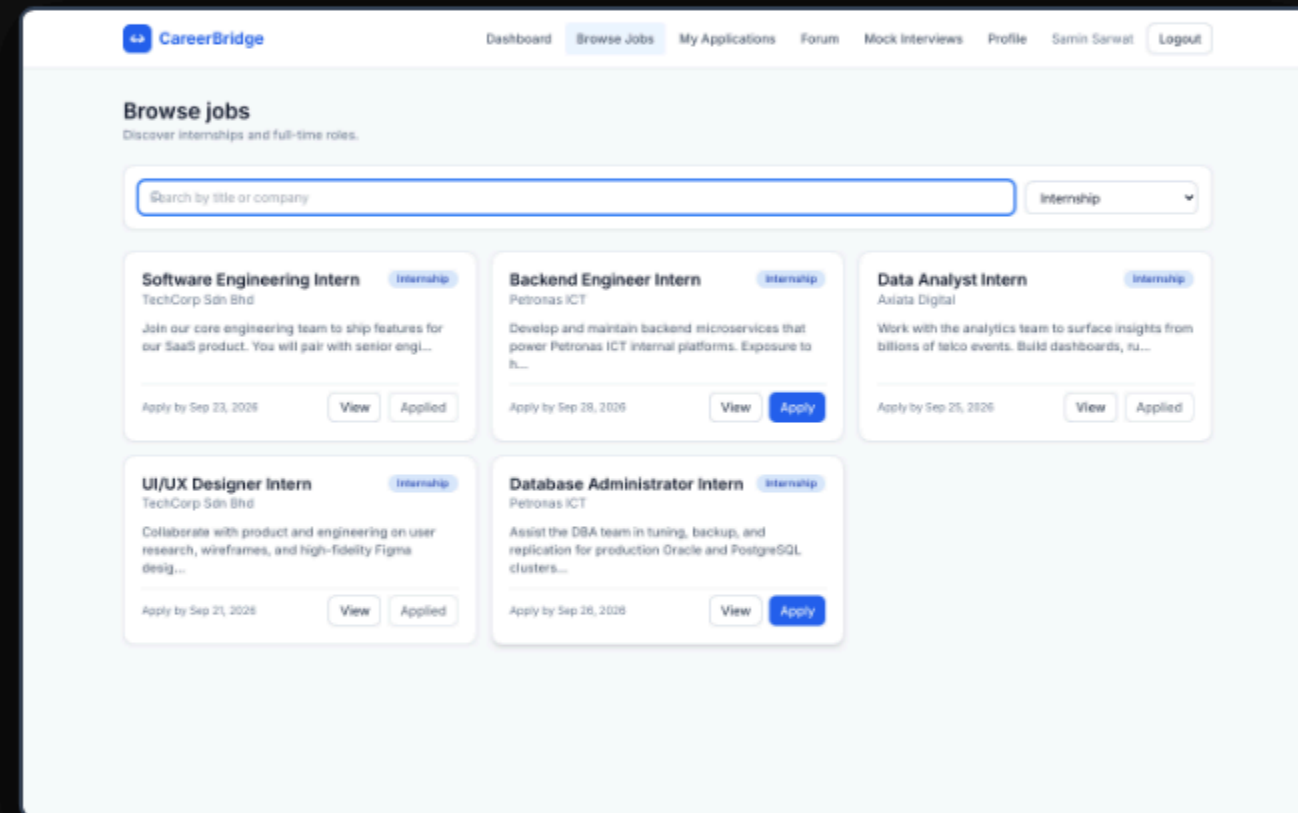


GET /api/jobs — loads all active jobs. Each card shows title, company, type badge, description, and deadline. Apply button disabled if already applied.

M2 Search & Type Filter

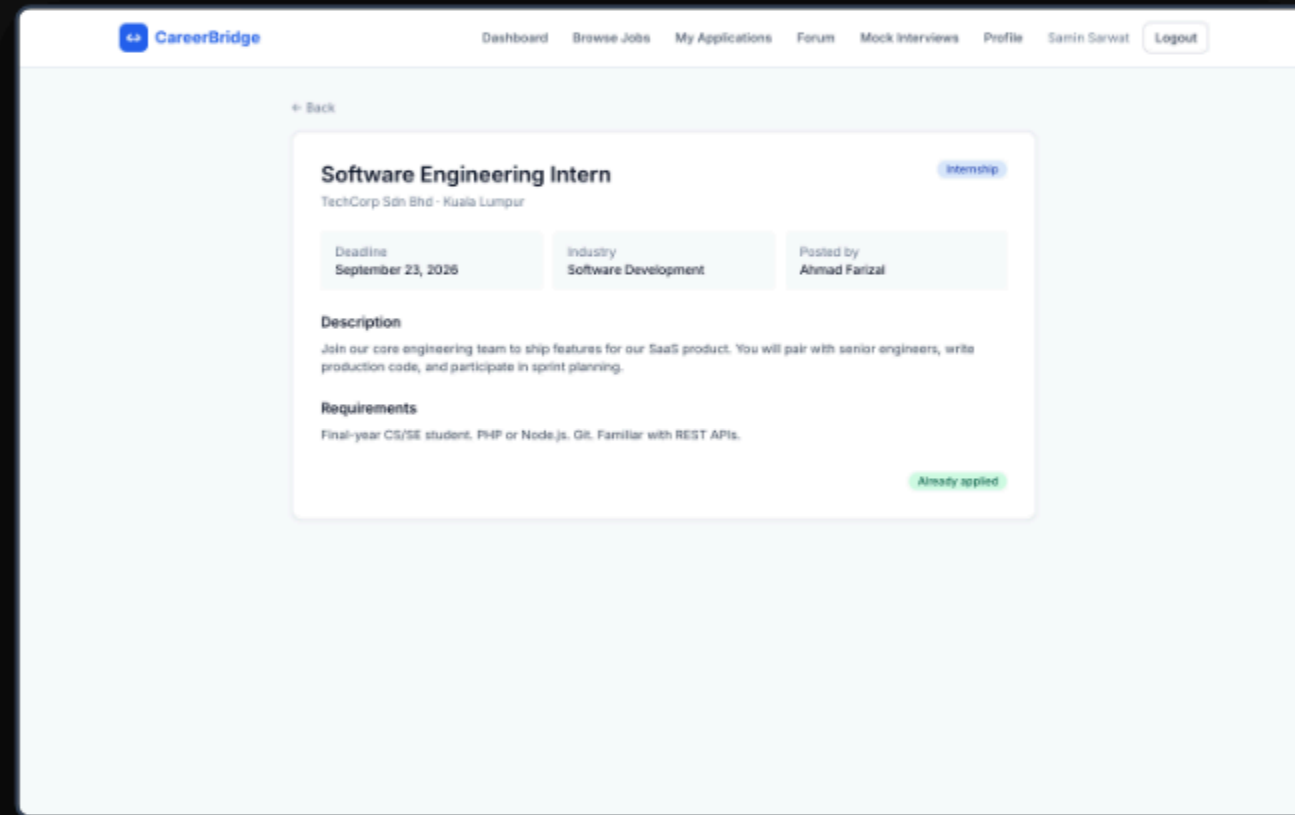


Search box with **400ms debounce** — filters by title or company name.



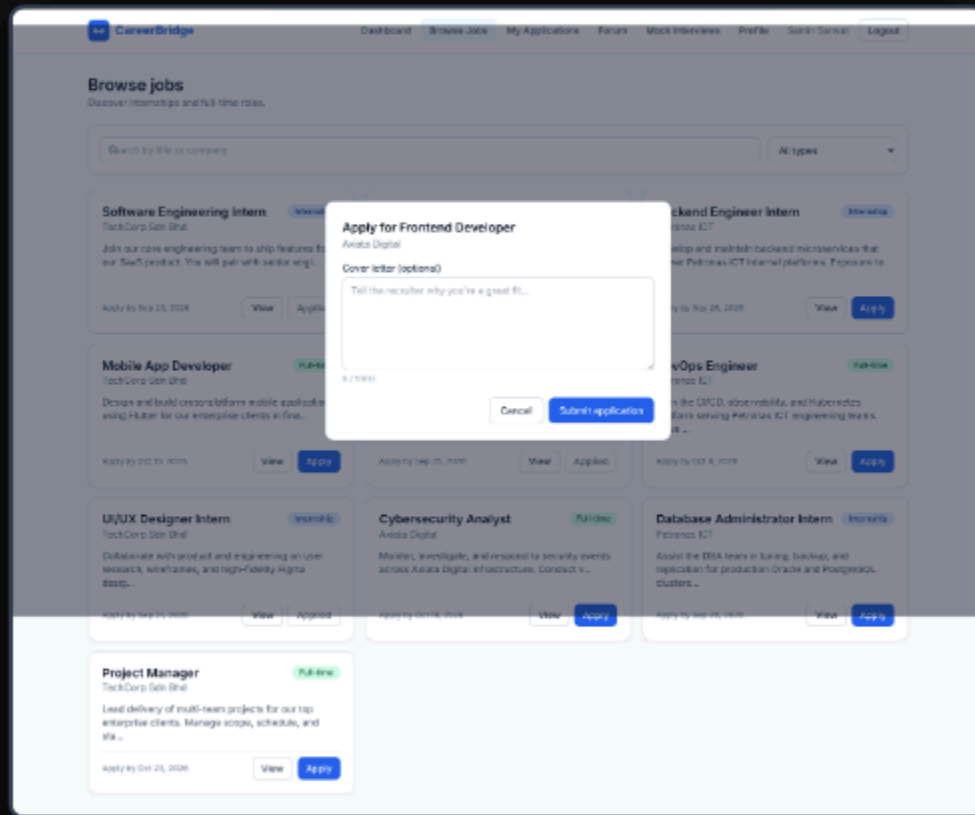
Dropdown filters by type: internship, full-time, part-time. Updates grid reactively.

M2 Job Detail View

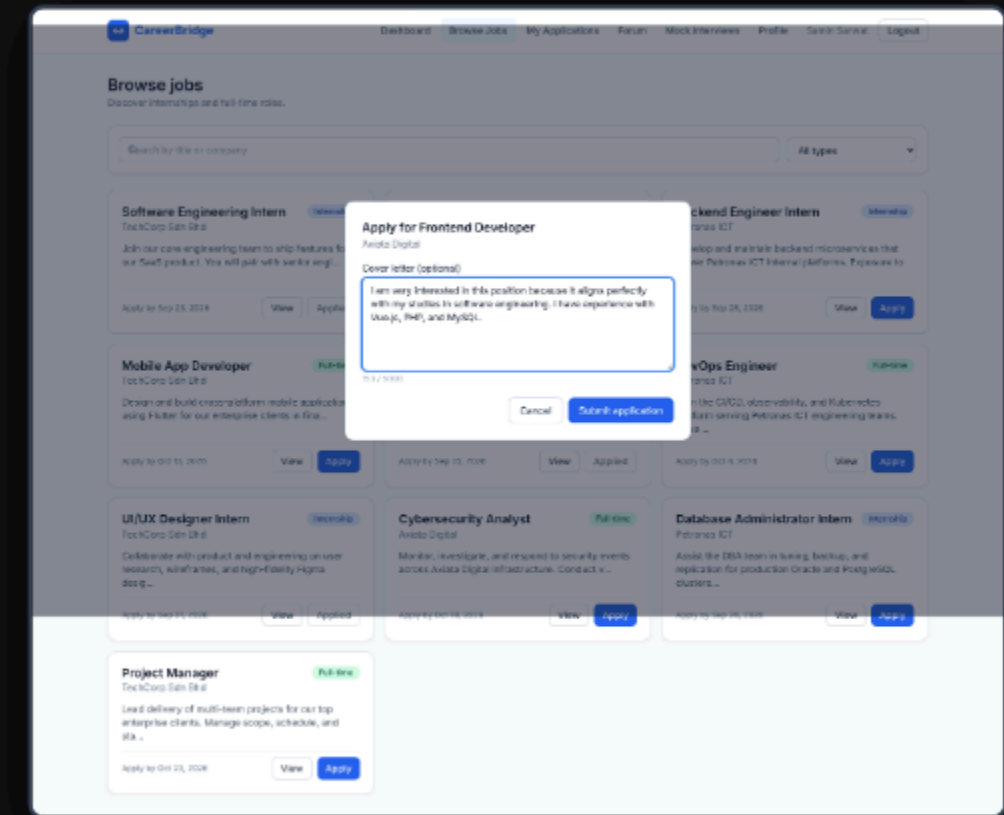


GET /api/jobs/{id} — Shows full description, requirements, deadline, company info. Backend uses JOIN to get company + posted_by name in one query.

M3 Apply for a Job

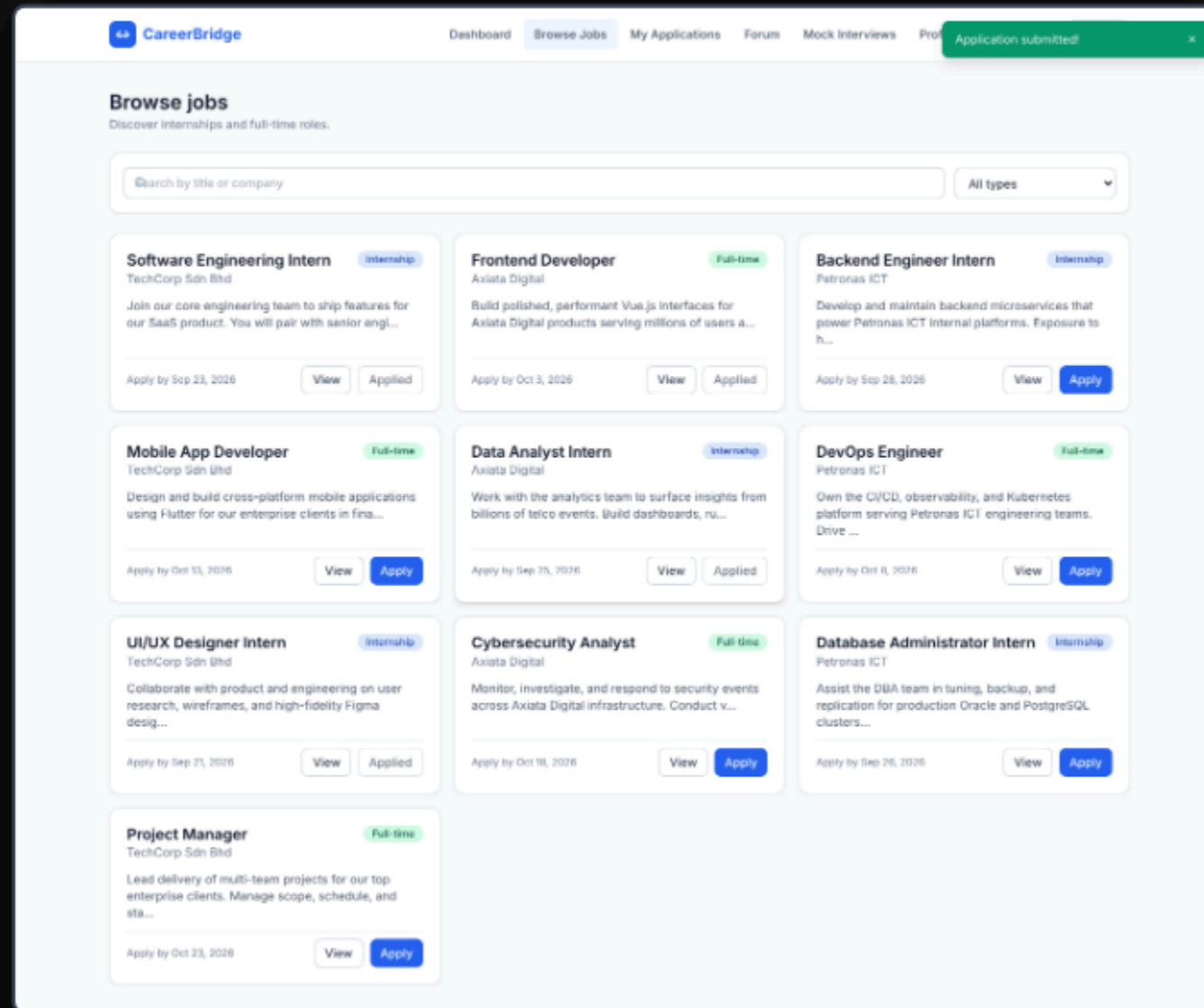


Click "Apply" opens a modal with optional cover letter.



Student writes a cover letter (max 5000 chars). Character counter shown.

M3 Application Submitted!



POST /api/applications — Backend checks: job exists, job active, not already applied (409 if duplicate). Toast confirms success. Apply button now disabled.

M3 My Applications (Student)

CareerBridge

DashboardBrowse JobsMy ApplicationsForumMock InterviewsProfileSamin SarwatLogout

My applications

Track every job you've applied for in one place.

All 4Pending 2Reviewed 0Accepted 1Rejected 1

JOB TITLE	COMPANY	APPLIED	STATUS	ACTIONS
Frontend Developer	Axiata Digital	Jun 25, 2026	Pending	Withdraw
Software Engineering Intern	TechCorp Sdn Bhd	Jun 25, 2026	Pending	Withdraw
Data Analyst Intern	Axiata Digital	Jun 25, 2026	Accepted	—
UI/UX Designer Intern	TechCorp Sdn Bhd	Jun 25, 2026	Rejected	—

GET /api/applications — Students see only their own. Tabs filter by status. Can withdraw pending applications only.

M3 My Applications — Pending Tab

CareerBridge

DashboardBrowse JobsMy ApplicationsForumMock InterviewsProfileSamin SarwatLogout

My applications

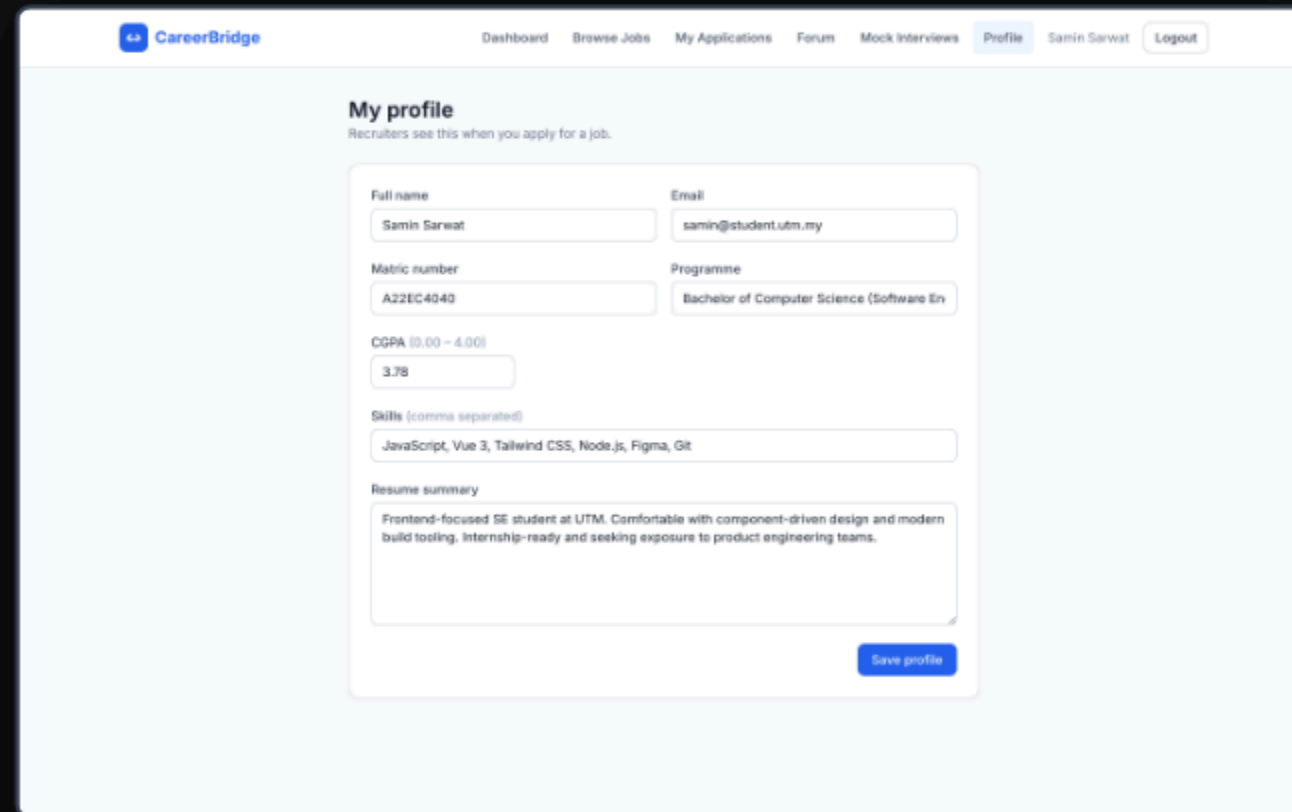
Track every job you've applied for in one place.

All 4Pending 2Reviewed 0Accepted 1Rejected 1

JOB TITLE	COMPANY	APPLIED	STATUS	ACTIONS
Frontend Developer	Axiata Digital	Jun 25, 2026	Pending	Withdraw
Software Engineering Intern	TechCorp Sdn Bhd	Jun 25, 2026	Pending	Withdraw

Filtering by "Pending" tab shows only applications awaiting review. Withdraw button available for these.

M6 Student Profile



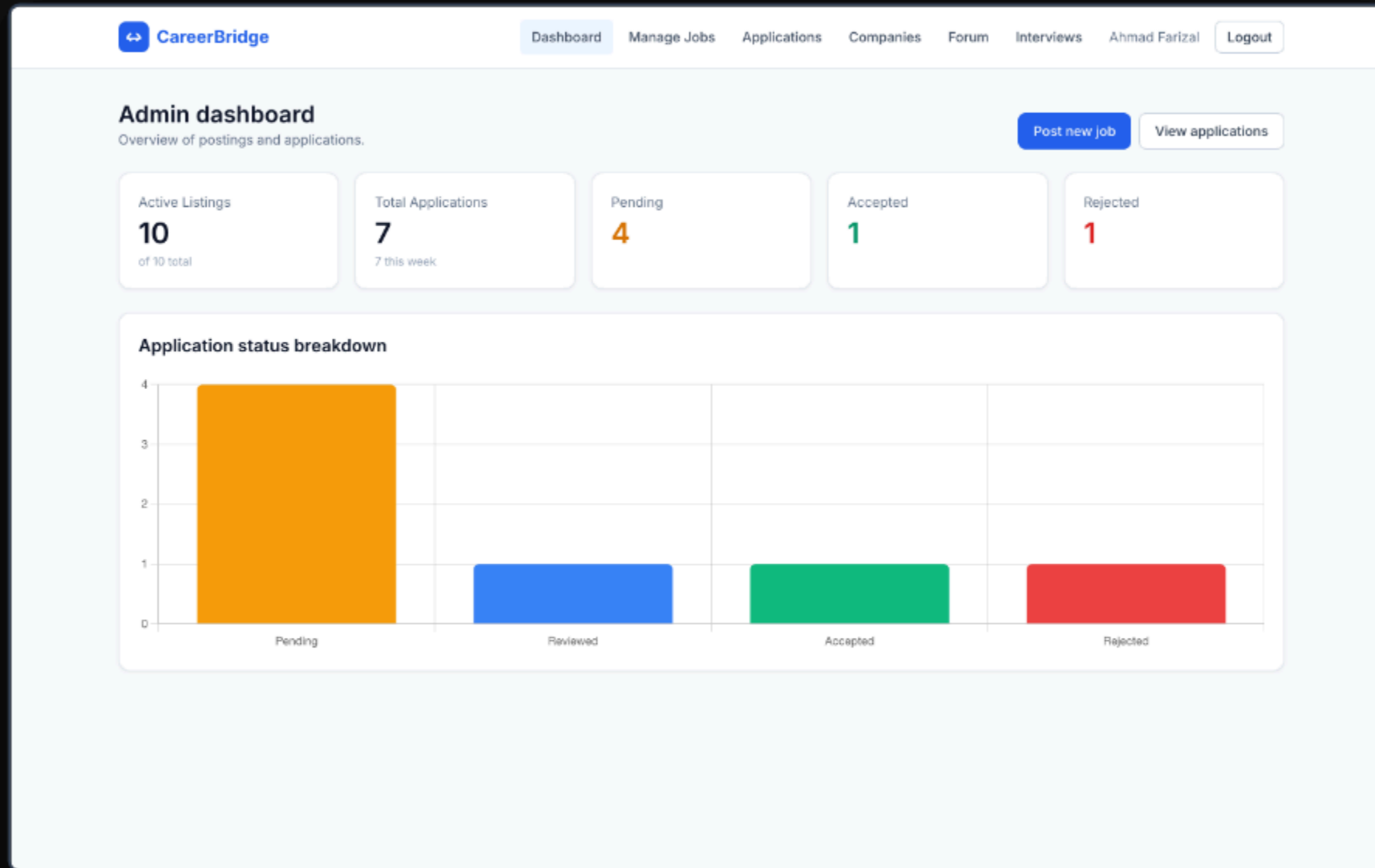
The screenshot shows the 'My profile' page on the CareerBridge website. The page has a navigation bar at the top with links: Dashboard, Browse Jobs, My Applications, Forum, Mock Interviews, Profile (active), Samin Sarwat, and Logout. The main content area is titled 'My profile' with a subtitle 'Recruiters see this when you apply for a job.' Below this is a form with the following fields:

- Full name: Samin Sarwat
- Email: samin@student.utm.my
- Matric number: A22EC4040
- Programme: Bachelor of Computer Science (Software En
- CGPA (0.00 – 4.00): 3.78
- Skills (comma separated): JavaScript, Vue 3, Tailwind CSS, Node.js, Figma, Git
- Resume summary: Frontend-focused SE student at UTM. Comfortable with component-driven design and modern build tooling. Internship-ready and seeking exposure to product engineering teams.

A 'Save profile' button is located at the bottom right of the form.

Matric number, programme, CGPA (0.00–4.00), skills, resume text. Uses PUT /api/profile (upsert — INSERT ... ON DUPLICATE KEY UPDATE).

M5 Admin Dashboard



GET /api/admin/analytics — Stat cards + Chart.js bar chart showing application status breakdown. Aggregates data from jobs and applications tables.

M2 Manage Jobs (Admin)

CareerBridge

DashboardManage JobsApplicationsCompaniesForumInterviewsAhmad FarizalLogout

Manage jobs

Create, edit, and retire job listings.

Add job

TITLE	COMPANY	TYPE	DEADLINE	STATUS	ACTIONS
Software Engineering Intern	TechCorp Sdn Bhd	Internship	Sep 23, 2026	Active	Edit Delete
Frontend Developer	Axiata Digital	Fulltime	Oct 3, 2026	Active	Edit Delete
Backend Engineer Intern	Petronas ICT	Internship	Sep 28, 2026	Active	Edit Delete
Mobile App Developer	TechCorp Sdn Bhd	Fulltime	Oct 13, 2026	Active	Edit Delete
Data Analyst Intern	Axiata Digital	Internship	Sep 25, 2026	Active	Edit Delete
DevOps Engineer	Petronas ICT	Fulltime	Oct 8, 2026	Active	Edit Delete
UI/UX Designer Intern	TechCorp Sdn Bhd	Internship	Sep 21, 2026	Active	Edit Delete
Cybersecurity Analyst	Axiata Digital	Fulltime	Oct 18, 2026	Active	Edit Delete
Database Administrator Intern	Petronas ICT	Internship	Sep 26, 2026	Active	Edit Delete
Project Manager	TechCorp Sdn Bhd	Fulltime	Oct 23, 2026	Active	Edit Delete

Admin sees all jobs with Edit and Delete (deactivate) buttons. Full CRUD interface. Status badge shows Active/Inactive.

M2 Create Job — Form with Validation

The screenshot shows the 'Manage jobs' page in the CareerBridge application. A 'New job' modal is open, displaying the following fields:

- Title:
- Company:
- Type:
- Deadline:
- Description:
- Requirements:

At the bottom of the modal are 'Cancel' and 'Create job' buttons. The background shows a list of job titles and their status (active, pending, etc.).

Empty form: title, company dropdown, type, deadline, description, requirements.

The screenshot shows the 'Manage jobs' page in the CareerBridge application. A 'New job' modal is open, displaying the following fields:

- Title:
- Company:
- Type:
- Deadline:
- Description:
- Requirements:

At the bottom of the modal are 'Cancel' and 'Create job' buttons. The background shows a list of job titles and their status (active, pending, etc.).

Frontend validates: required fields, future deadline. Backend validates again.

M3 Manage Applications (Admin)

CareerBridge

DashboardManage JobsApplicationsCompaniesForumInterviewsAhmad FarizalLogout

Applications

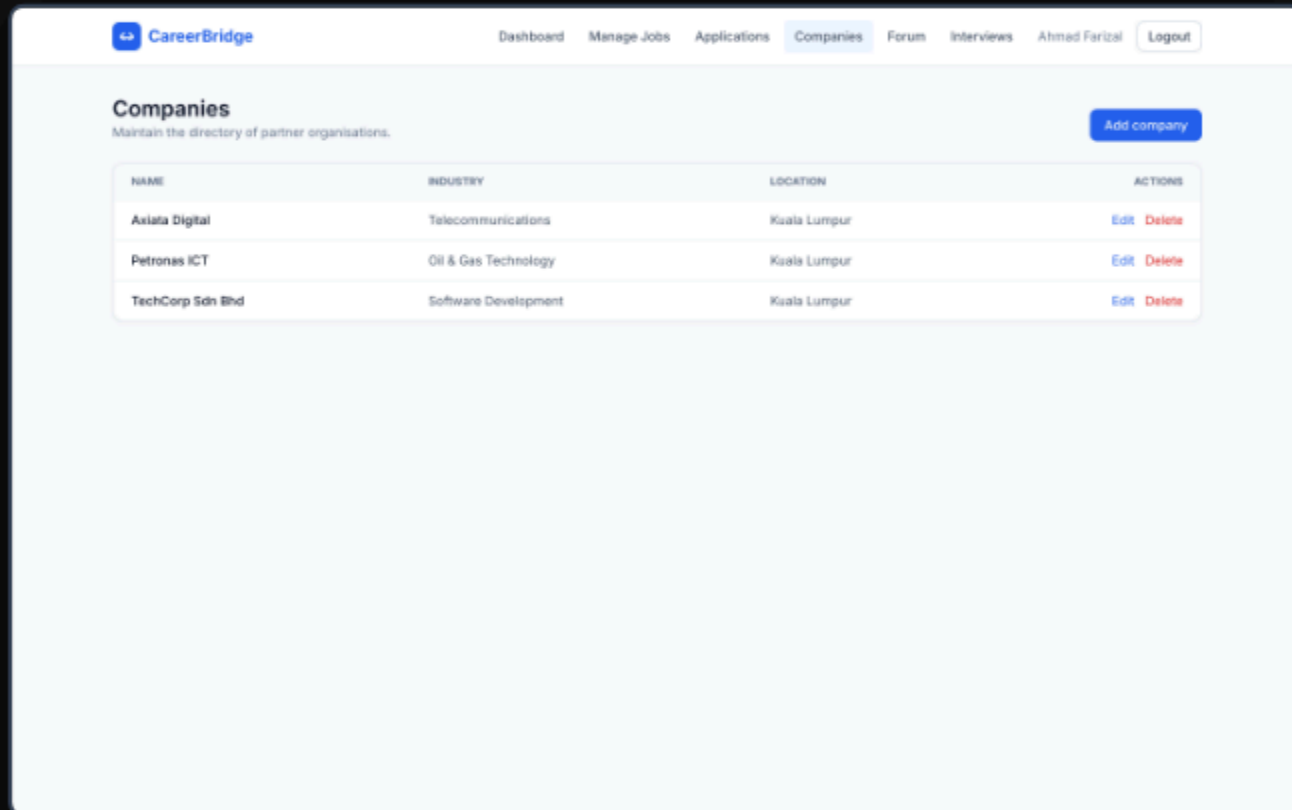
Move applications through the hiring pipeline.

All 7Pending 4Reviewed 1Accepted 1Rejected 1

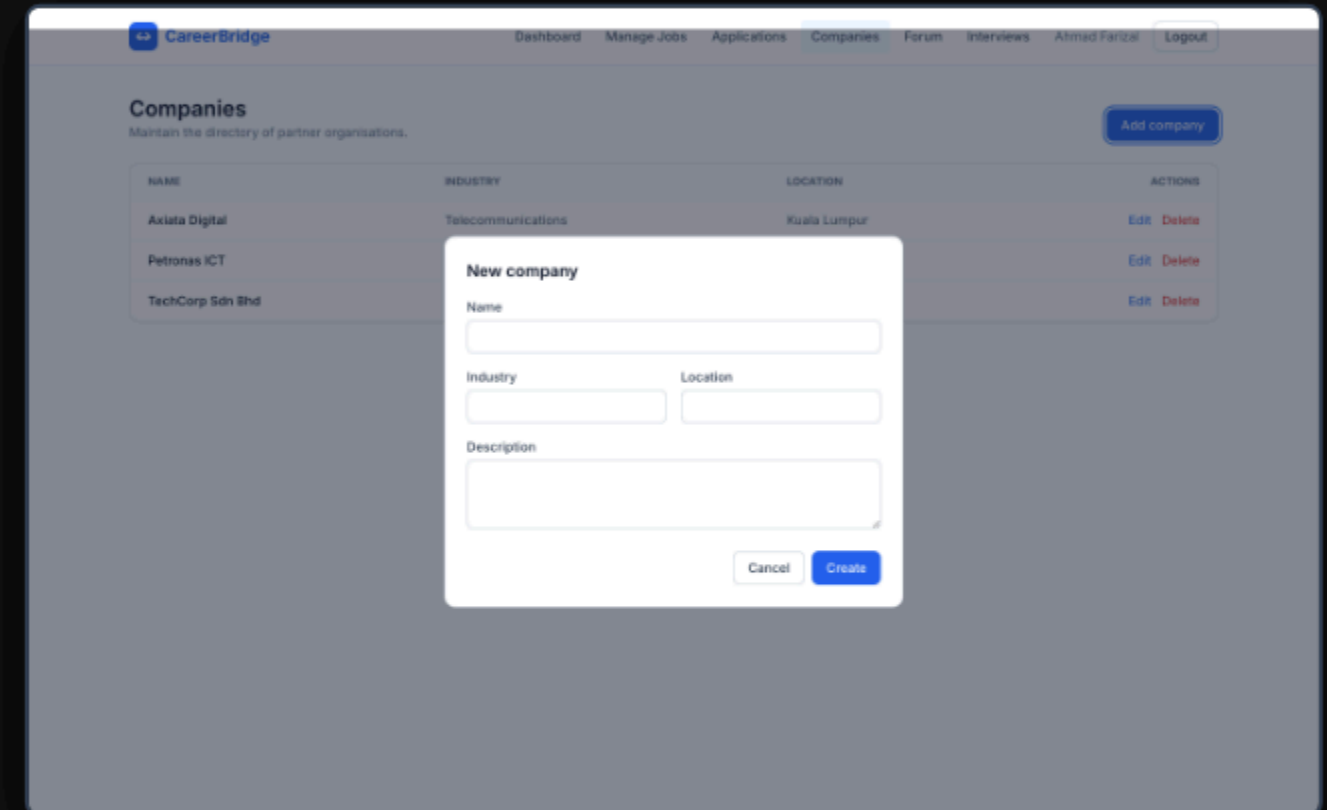
APPLICANT	JOB	COMPANY	APPLIED	CURRENT	UPDATE STATUS
Samin Sarwat samin@student.utm.my	Frontend Developer	Axiata Digital	Jun 25, 2026	Pending	pending
Samin Sarwat samin@student.utm.my	Software Engineering Intern	TechCorp Sdn Bhd	Jun 25, 2026	Pending	pending
Mohammad Areeb areeb@student.utm.my	Software Engineering Intern	TechCorp Sdn Bhd	Jun 25, 2026	Pending	pending
Mohammad Areeb areeb@student.utm.my	Backend Engineer Intern	Petronas ICT	Jun 25, 2026	Reviewed	reviewed
Samin Sarwat samin@student.utm.my	Data Analyst Intern	Axiata Digital	Jun 25, 2026	Accepted	accepted
Samin Sarwat samin@student.utm.my	UI/UX Designer Intern	TechCorp Sdn Bhd	Jun 25, 2026	Rejected	rejected
Mohammad Areeb areeb@student.utm.my	Database Administrator Intern	Petronas ICT	Jun 25, 2026	Pending	pending

Admins see ALL applications. Dropdown changes status via PUT /api/applications/{id}/status. Uses **optimistic update** — UI updates instantly, rolls back on error.

M4 Manage Companies (Admin)



Admin sees all companies with Edit and Delete buttons.



Create company form: name, industry, location, description.

Companies are linked to jobs via foreign key. Cannot delete a company that has jobs attached (returns 400 error).

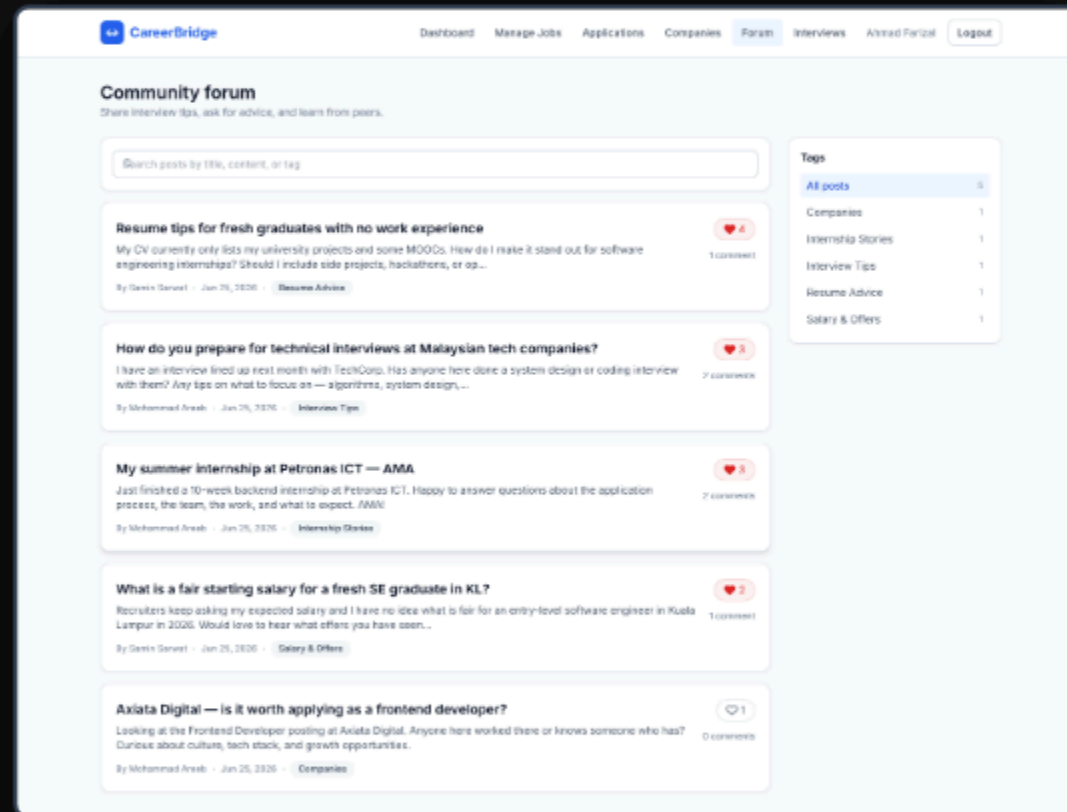
M8 Mock Interviews (Admin)

The screenshot shows the 'Manage mock interviews' page in the CareerBridge admin dashboard. The page has a header with the CareerBridge logo and navigation links: Dashboard, Manage Jobs, Applications, Companies, Forum, Interviews, Ahmad Farizal, and Logout. Below the header, the page title 'Manage mock interviews' is followed by a subtitle 'Open new availability slots and evaluate completed sessions.' and an '+ Add slot' button. The main content area features a table with two tabs: 'Slots' (selected) and 'Bookings'. The table has four columns: SCHEDULED AT, INTERVIEWER, STATUS, and ACTIONS. It lists five interview slots, with the first one booked and the others open.

SCHEDULED AT	INTERVIEWER	STATUS	ACTIONS
Sun, Jun 28, 2026, 10:37 PM	Ahmad Farizal	Booked	—
Thu, Jul 2, 2026, 10:37 PM	Ahmad Farizal	Open	Delete
Sun, Jul 5, 2026, 10:37 PM	Ahmad Farizal	Open	Delete
Thu, Jul 9, 2026, 10:37 PM	Super Admin	Open	Delete
Thu, Jul 16, 2026, 10:37 PM	Super Admin	Open	Delete

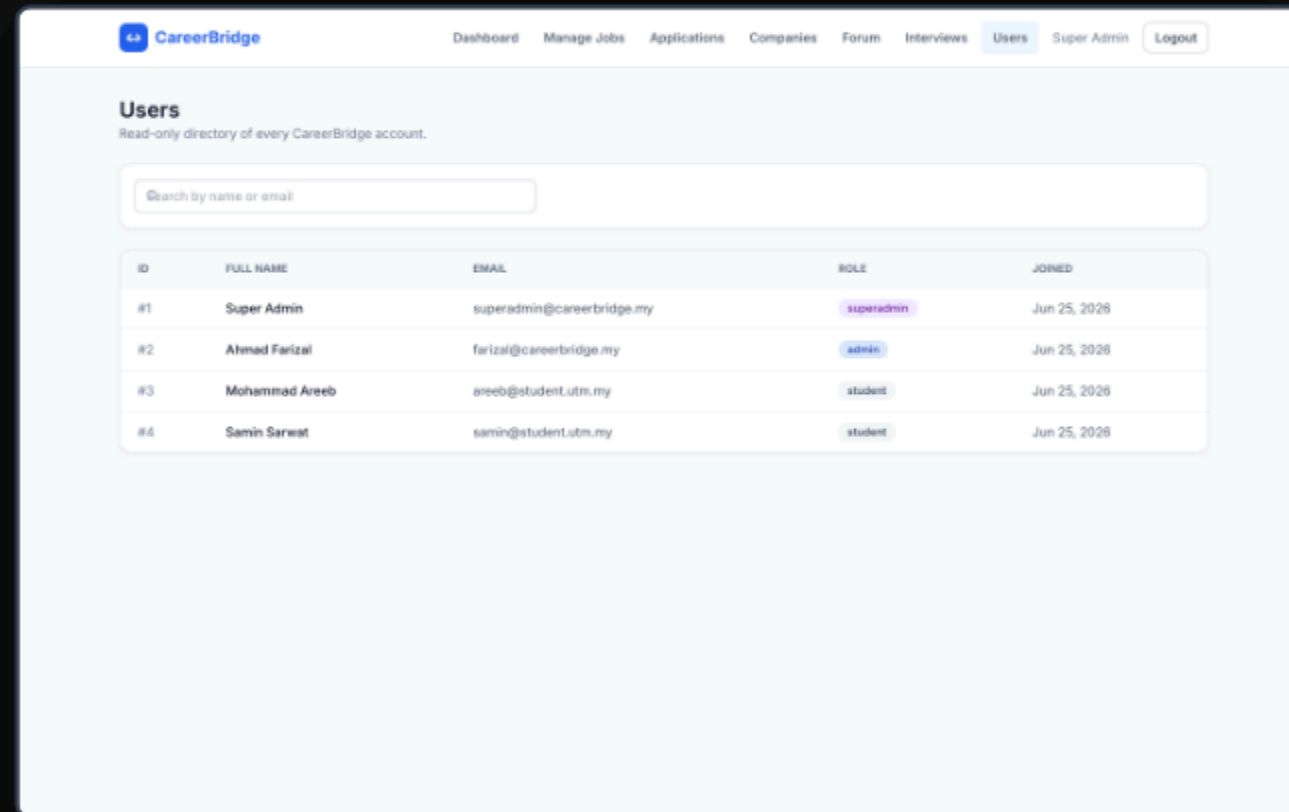
Admins create interview slots with date/time. Students book available slots. Admins evaluate with scores and feedback text.

M7 Forum / Community



Students post questions and discussions with tags. Other students can comment and like posts. Search and filter by tag.

M1 Admin Users (Super Admin Only)



The screenshot shows the 'Users' management page in the CareerBridge admin interface. At the top, there's a navigation bar with links to Dashboard, Manage Jobs, Applications, Companies, Forum, Interviews, Users (active), Super Admin, and Logout. Below the navigation bar, the page title 'Users' is followed by a subtitle 'Read-only directory of every CareerBridge account.' A search bar labeled 'Search by name or email' is positioned above a table. The table lists four users with columns for ID, Full Name, Email, Role, and Joined. The roles are displayed as colored badges: 'superadmin' in purple, 'admin' in blue, and 'student' in grey.

ID	FULL NAME	EMAIL	ROLE	JOINED
#1	Super Admin	superadmin@careerbridge.my	superadmin	Jun 25, 2026
#2	Ahmad Farizal	farizal@careerbridge.my	admin	Jun 25, 2026
#3	Mohammad Areeb	areeb@student.utm.my	student	Jun 25, 2026
#4	Samin Sarwat	samin@student.utm.my	student	Jun 25, 2026

GET /api/admin/users — Only superadmin role can access. Read-only directory of all accounts with role badges.

Security Features

Backend Security

- **PDO Prepared Statements** — all queries use `:placeholder` syntax, preventing SQL injection
- **JWT Role Gating** — `JwtMiddleware` on protected routes
- **Bcrypt Passwords** — `password_hash()` with cost 12
- **Ownership checks** — students can only access their own data
- **Soft delete** — jobs deactivated, not erased (audit trail)
- **Duplicate prevention** — UNIQUE constraint on (job_id, user_id)
- **Referential integrity** — can't delete company with linked jobs

Frontend Security

- **Route guards** — router checks role before showing page
- **Auto-logout on 401** — Axios interceptor clears token
- **Client-side validation** — all forms validated before API call
- **Confirm dialogs** — before destructive actions (delete, withdraw)
- **JWT in localStorage** — attached to every request via interceptor
- **Disabled buttons** — Apply disabled if already applied

HTTP Status Codes Used

200

OK — GET, PUT, DELETE success

201

Created — POST success

400

Bad Request — validation failed

401

Unauthorized — bad/missing JWT

403

Forbidden — wrong role or not owner

404

Not Found — resource missing

409

Conflict — already applied / duplicate

Complete API — All Endpoints

M1 Auth & Users

- POST /api/auth/login — Login
- POST /api/auth/register — Register
- GET /api/admin/users — Super admin only

M2 Jobs

- GET /api/jobs — List (public)
- GET /api/jobs/{id} — Detail (public)
- POST /api/jobs — Create (admin)
- PUT /api/jobs/{id} — Update (admin)
- DELETE /api/jobs/{id} — Soft delete (admin)

M3 Applications

- GET /api/applications — Role-aware list
- POST /api/applications — Apply (student)
- PUT /api/applications/{id}/status — Update (admin)
- DELETE /api/applications/{id} — Withdraw (student)

M4 Companies

- GET /api/companies — List (public)
- POST /api/companies — Create (admin)
- PUT /api/companies/{id} — Update (admin)
- DELETE /api/companies/{id} — Delete (admin)

M5 Analytics

- GET /api/admin/analytics — Dashboard stats (admin)

M6 Profile

- GET /api/profile — View own profile
- PUT /api/profile — Upsert profile

M7 Forum

- GET /api/posts — List posts
- POST /api/posts — Create post
- POST /api/comments — Add comment
- POST /api/posts/{id}/like — Toggle like

M8 Interviews

- GET /api/interview-slots — List slots
- POST /api/interview-slots — Create (admin)
- POST /api/mock-interviews/book — Book (student)
- PUT /api/mock-interviews/{id} — Evaluate (admin)

Team CHAMPION — Module Ownership

Module	Name	Owner	Key Feature
M1	Auth & Users	Areeb	JWT login, role middleware, shared infrastructure
M2	Job Management	Samin	Full CRUD, soft delete, search + filter
M3	Application Tracking	Samin	Apply, withdraw, role-aware listing, status flow
M4	Company Management	Mariam	CRUD, referential integrity with jobs
M5	Reporting & Analytics	Mariam	Dashboard stats, Chart.js visualization
M6	Student Profile	Areeb	Upsert pattern, CGPA validation
M7	Forum / Community	Monika	Posts, comments, likes, tag filtering
M8	Mock Interviews	Monika	Slot booking, evaluation, scores

System at a Glance

8

Modules

25+

API Endpoints

10

Database Tables

3

User Roles

15+

Vue Views

8

Backend Controllers

Built with: Vue 3 + Vite + Tailwind CSS + PHP Slim 4 + MySQL + JWT + PDO + Chart.js

Thank You

CareerBridge — Student Internship & Job Application Management System

Team CHAMPION

Vue 3 · PHP Slim 4 · MySQL · JWT Authentication · PDO Prepared Statements · Chart.js

Any questions?

